NR–REX NETWORK INTEGRATION PROJECT EXECUTION PLAN

Client: National Railways (NR)

PREPARED BY GROUP 03

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NR–ReX Network Integration Project
Project Execution Plan

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Kavish Gunasekara (Project Manager), Darren Lim (Technical Lead), Sarik (Business Analyst)

The following two paragraphs should be included in the documents where these templates have been used as a basis for development.

This document has been derived from a template prepared by the Department of Premier and Cabinet, Tasmania. The structure is based on several methodologies as described in the *Tasmanian Government Project Management Guidelines*.

DOCUMENT ACCEPTANCE and RELEASE NOTICE

This is Version 1.0 (Final) NR–ReX Network Integration Project Execution Plan.

The Project Execution Plan is a managed document. For identification of amendments each page contains a release number and a page number. Changes will only be issued as complete replacement. Recipients should remove superseded versions from circulation. This document is authorised for release once all signatures have been obtained.

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1 Introduction

1.1 Document Purpose

The Project Execution Plan (PEP) showcases how the National Railways (NR) – Regional Express (ReX) Network Integration Project will be carried out during its 20-month period.

The PEP converts the approved Project Charter (Stage 1) and Project Plan (Stage 2) into a practical, sequential guide for daily execution.

This document is used by the entire team to coordinate activities, manage progress, and control quality, cost, and risk: While being maintained by the Project Manager.

It acts as the project's operational roadmap, ensuring that all team members (New or already existing) acknowledge and understand the tasks, communication methods, testing approach, and acceptance processes to ensure that the necessary outcomes are met.

It will be reviewed and updated throughout the project life cycle to reflect changes approved by the sponsor or steering committee.

1.2 Intended Audience

The PEP is intended for:

Executive Sponsor (Ochini) – for approval of milestones and observing progress.

Business Sponsor (Michael) – for oversight of integration outcomes, costs, and efficiency targets.

Project Team (Kavish Gunasekara, Sarik, Darren) – for reference during daily operations, reporting, testing, and documentation where needed.

Stakeholders from NR and ReX – consisting of depot managers, scheduling staff, and IT support who are impacted by the novel integrated systems.

Assumptions

- Readers are already familiar with the basic knowledge of the NR – ReX integration objectives and core project management principles.

- The document will be updated accordingly through several iterations as each phase is concluded and completed.

- Version control details and amendment history will document any changes made.

1.3 Project Outputs

Output	Description
Integrated Fleet Management System	Consolidated platform for NR and ReX depots where vehicle tracking, scheduling, and maintenance can be managed.
Digital Tracking and Route Optimization Tool	Real-time GPS solution to improve delivery times and reduce logistics costs where possible.
Standardized Operating Procedures (SOPs)	Standardized processes and policies for fleet operations and communication flows.
Staff Training and Change Management Program	Structured training modules to ensure smooth adoption of new systems.
Final Integration Report and Maintenance Framework	Comprehensive handover package including support, disaster recovery, and lessons learned.

1.4 Scope of Work

Within Scope	Outside Scope
Alignment of NR and ReX fleet management systems.	Procurement of new trains or vehicles.
Integration of digital tracking and communication tools.	Major infrastructure construction or facility upgrades.
Data migration and testing of operational systems.	Corporate restructuring or HR policy changes.
Staff training and change management programs.	Non-transport divisions or subsidiaries of NR.
Pilot and nationwide rollout of integrated system.	International operations outside Australia.

2 Management Plan

2.1 Management

The project will be managed by Project Manager (Kavish). He is responsible to Executive Sponsor (Ochini) and Business Sponsor (Michael) for the delivery of the agreed project outputs. Project Manager (Kavish) is accompanied by and assigned Business Analyst (Sarik) and Technical Lead (Darren)within a team. Decisions involving operational issues and critical issues are referred to Business Sponsor (Michael) while strategic changes to the scope of the project comes under the examination of Executive Sponsor (Ochini).

2.1.1 Introduction

The Stage 2 Project Plan forms the basis of operational management within the project. The project manager has multiple responsibilities, including day-to-day coordination and progress tracking, while communicating with sponsors regularly. This can be seen within weekly meetings to ensure the project is on track and provides opportunities to address any changes.

2.1.2 Sub-Project Management

Within the project phases, there are sub-phases (Data migration, training of staff members, and system integration) within them. These undergo the same management as the main phases. There are no sub-phases done independently or separately to the main phases or the project.

2.1.3 Reference Groups

A steering committee acts as the primary reference group within the project.

- Members: Executive Sponsor, Business Sponsor, Project Manager, Technical Lead.
- Objectives: Review progress, approval of milestones, resolve escalated issues, and validate project alignment with organizational strategy.
- Frequency of meetings: At each major milestone and/or monthly.

2.1.4 Consultants

External consultants may be utilized for security audits and system integration. Project Manager will receive their report and receive specialist input on risk mitigation/technical configurations during the build and pilot phases.

2.1.5 Working Parties

A temporary change management working group consisting of depot supervisors and training leads from NR and ReX will coordinate rollout readiness and feedback collection during pilot implementation.

2.2 Status Reporting

Status reporting makes sure that accountability and transparency are maintained throughout the 20-month project.
Frequency & Responsibility:

- Weekly Reports – prepared by the Project Manager summarising task progress, issues, and next steps.
- Monthly Reports – consolidated for the Steering Committee, including milestone achievements, financial status, and risk updates.
- Quarterly Summaries – shared with sponsors and key stakeholders outlining overall performance and ROI projections.

Content of Reports:

- Current progress against milestones and budget.
- Planned activities for the next reporting period.
- Issues log and required decisions.

- Updated risk register with mitigation actions.

For accessibility and version control, all reports are stored on SharePoint.

2.3 Risk Management

Risk management practices happen continuously at the start of each project phase and monthly thereafter.

The Steering Committee reviews significant updates. The Project Manager, Technical Lead, and Business Analyst take over review of minor updates.

2.3.1 Risk Assessment

- Frequency: At the beginning of each project phase and before milestone sign-off.
- Participants: Project Manager, Business Sponsor, Team Leads, and NR/ReX representatives.
- Documentation: Results recorded in the Risk Register (Appendix B).
- Monitoring: Risks reviewed weekly; effectiveness of mitigation tracked through performance metrics.
- Approval: Mitigation strategies requiring cost or scope change must be endorsed by the Steering Committee.

2.3.2 Failure to Deliver

If a milestone is not delivered by more than one month from the scheduled date, the Project Manager will provide a recovery plan to the Business Sponsor detailing changed timelines, risk impacts and resource requirements. The Steering Committee will approve or change the corrective action accordingly.

2.3.3 Acceptance and Review Periods

All reviews and approvals of outputs will be completed within 10 working days of submission to the Business Sponsor. Delays beyond this period will prompt schedule reassessment and updated timelines.

2.3.4 Issues

Issues that cannot be resolved internally will be escalated in writing to the Business Sponsor using an Open Issues Form. The recipient must acknowledge and document resolution within five working days.

2.3.5 Non-availability of Resources

If critical personnel or equipment become unavailable, the Project Manager will notify the Steering Committee and reschedule dependent activities. Resource reallocation or temporary replacements will be approved by the Business Sponsor.

2.4 Provision of Facilities and Equipment

The Project Team will leverage an office provided by the University and use MS Teams, SharePoint along with NR and ReX's tool for testing.

The IT Division of NR manages technical environments such as secure VPN access, test servers and VMs. The standard maintenance procedures of Stage 2 will apply to all equipment.

2.5 Skills and Resource Requirements

2.5.1 Skills and Resources

The project requires the following key roles and capabilities:

- Project Manager: Leadership, planning, communication, and risk management.
- Business Analyst: Requirements mapping, documentation, and testing coordination.
- Technical Lead: System design, integration, and troubleshooting.
- Quality Officer: Independent verification and quality checks.
- Trainers / Change Manager: Training development and user adoption.

Most resources are part-time student team members, supplemented by NR staff for real-world input. Absences are mitigated through shared task documentation and meeting minutes.

2.5.2 Training

Training includes:

- Internal workshops for the project team on Microsoft Project, SharePoint, and Jira.
- Staff training sessions at NR and ReX depots to ensure proficiency in the integrated platform.
- Refresher courses during the rollout phase to sustain adoption.

2.6 Configuration Management

Configuration management controls changes to the project scope, outputs, and documentation.

2.6.1 Change Control

Change requests are logged in SharePoint and reviewed weekly by the Project Manager.
The process ensures:

- Impact analysis before approval.
- Steering Committee authorisation for major changes.
- A complete audit trail for traceability.

2.6.2 Problem Reporting and Resolution

Jira is used to log in to any problems that occurred during testing/development.
The problem is attached to a priority, owner, and resolution deadline, along with status reports in weekly meetings.

2.6.3 Incident Reporting

The Incident Form will be created for incidents such as data loss, unauthorised access, or major technical errors that can be reported to the Technical Lead and Business Sponsor and escalated directly to the Technical Lead and Business Sponsor.

2.6.4 Issues Management

The project team is involved with small issues; problems left unresolved are reported to the Business Sponsor or Steering Committee as described in Section 2.3.4.

2.7 Confidentiality

All of the team members, contractors, and stakeholders are required to maintain confidentiality in regard to project information and NR/ReX operational data.
All collaboration agreements also contain non-disclosure clauses and data sharing follows NR's information-security policies.

2.8 Output Review and Acceptance

The Project Manager reviews each delivery and quality validation of each deliverable before presenting it to the Business Sponsor.

Acceptance criteria include:

- Conformance to functional and quality specifications.
- Documentation completeness.
- Approval by both the Business Sponsor and Executive Sponsor prior to milestone closure.

2.9 Updating This Plan

Each main phase of the project will be subjected to a review and only updated if found necessary. Any big changes in how the project is scheduled, managed or resourced will be limited and renewed only after the sponsor approves of the changes. Minor changes (E.g. Staff and tasks) will be recorded in monthly progress reports.

3 Quality Plan

3.1 Introduction

The management/maintenance of quality assurance within the project plan is seen and developed in the Project Quality Plan. This is vital to ensure that the products from the project are reliable, functional and implemented when needed. These can be seen in: Proven project management/development methodologies, continuous monitoring/review of progress, effective management of change requests/problems, formal review/acceptance processes for

all outputs.

3.2 Methodologies and Standards

Adopted Methodologies and Standards:

- Project Management Framework: Based on the PMBOK (Project Management Body of Knowledge) principles, allowing for the structured planning, execution, monitoring, and closure of all phases.

- Quality Management: Guided by ISO 9001:2015 standards, focusing on continuous improvement and documentation control.

- Software Development & Integration: Follows a Waterfall Methodology with milestone-based checkpoints (Initiation → Design → Build → Test → Implement → Close).

- Documentation Standards: Use of consistent templates for charters, plans, reports, and registers to ensure readability and traceability.

- Change and Risk Management: Governed by a controlled register system using SharePoint and Jira.

3.3 Development Environment

The development environment is the enabler for integration between NR and ReX systems.
Environment Components:

- Tools & Software:
    - Microsoft Project for scheduling and Gantt management.
    - Jira for issue tracking.
    - SharePoint for document collaboration.
    - SQL database for migration and testing.
    - Microsoft Windows as an operating platform

- Testing Environment: A sandbox network provided by NR's IT division for simulation of data migration and tracking functions.

3.4 Inspection, Measuring, and Test Equipment

The project will utilise both software-based and manual tools to ensure quality assurance across deliverables.

Inspection and Testing Tools:
- Automated test scripts for verifying data migration accuracy.
- System monitoring dashboards for tracking performance and uptime.

- Security and penetration testing tools for validating data protection compliance.
- Manual test checklists for user acceptance and functional validation.

3.5 Development Cycle

The project undergoes the Waterfall Development Cycle with milestones at each phase.

Phases

1. Initiation & Planning: Define objectives, scope, and governance.
2. Requirements & Design: Document integration needs and system architecture.
3. Build & Testing: Develop integration modules, perform UAT and quality validation.
4. Implementation: Deploy solutions across pilot and full rollout stages.
5. Closure: Conduct evaluation, final documentation, and sign-off.

3.6 Outputs to Be Developed

Output	Description
Integration Framework	Standardised operational and digital processes linking NR and ReX systems.
Data Migration Pack	Cleansed and validated datasets migrated into NR's central database.
Tracking & Scheduling System	Unified software platform for fleet tracking and delivery optimisation.
Training & Change Material	Manuals, training modules, and communication templates for staff adoption.
Project Documentation	Includes the Project Charter, Plan, Execution Plan, Closure Report, and Lesson Learned Report.
Quality & Testing Reports	Verification results for functional, integration, and performance testing.

3.7 Project Evaluation

Project evaluation measures the success of both process and product quality.

Evaluation will occur at the end of each project phase and after final delivery.

3.8 Records

3.8.1 Record Keeping

All project records will be kept within Microsoft SharePoint.

Records that should be maintained are of the following below:

- Project Management Documents (Charter, Plan, Execution Plan).
- Risk, Issue, and Change Registers.

- Test Results, Inspection Logs, and QA Reports.
- Training Attendance and Evaluation Logs.
- Meeting Minutes and Approval Sign-offs.

Version history, author, and date of approval information will be attached to every record.

3.8.2 Records Required by Business Owner

If asked by the Business Sponsor, all requested records are provided to them promptly. Records will be shared with them formally during reviews. Furthermore, they have access to archives of the records project after the project is finished.

3.8.3 Retention of Records

All project records will be securely stored and archived for 5 years when project is completed.

4 Development Plan

The NR-ReX Network Integration Project includes the documentation and process for creating a number of outcomes related to the interoperability of their networks such as integrated tracking systems, harmonized standard operating procedures, necessary data movements packages, testing documentation, as well as the required training materials.

The how, with whom, with what and how these outputs will be regulated and delivered to quality, on time and to specification are all detailed in this Development Plan.

4.1 Design and Development Activities

All design and development activities for project outputs will follow a structured and traceable process:

1. Requirements Confirmation – Review business and functional requirements defined in Stage 2 (Project Plan).

2. System & Process Design – Create high-level architecture diagrams, data-flow maps, and SOP drafts.

3. Development / Configuration – Configure integration systems, set up data-migration scripts, and prepare testing environments.

4. Internal Review & Validation – Conduct peer reviews and system walkthroughs to ensure compliance with standards.

5. User Acceptance Testing (UAT) – Validate the system and processes with NR and ReX staff.

6. Final Approval & Distribution – Obtain sponsor sign-off before rollout and document final versions in the Output Register.

Primary documentation outputs:

- Functional Requirements Specification (FRS)
- System Design Specification (SDS)
- Test Plan and UAT Reports
- Training Manuals and User Guides
- Maintenance Procedures and Handover Documents

4.2 Organisation and Staffing

Role	Key Responsibilities	Involvement
Project Manager (Kavish Gunasekara)	Oversees all development progress, ensures milestone delivery, approves design documentation	Full-time
Technical Lead (Darren)	Designs system architecture, oversees data migration and integration testing	Full-time
Business Analyst (Sarik)	Validates functional requirements, supports SOP development and testing	Part-time
Integration Engineers / Vendors	Provide specialist technical support for APIs and software tools	Contract
Change Manager (NR Staff)	Reviews training materials and ensures operational readiness	Part-time

4.3 Design Methodology

The project will use a Waterfall development methodology in the steps of requirements, design, build, test, and deploy.

This process should help ensure that the necessary validation of each step is done and traceability and rework are minimized.

Methodology References:

- PMBOK (Project Management Body of Knowledge) principles
- NR internal IT standards and security protocols
- ISO 9001 Quality Management principles

4.4 Design Input

Source	Description
Project Charter (Stage 1)	Defines project goals, assumptions, and scope boundaries
Project Plan (Stage 2)	Details schedule, WBS, milestones, and resource allocations
Requirements Analysis Report	Documents business needs and system functionality
Existing NR & ReX Systems Data	Source data used for integration and migration design

Standards and Policies	NR security, privacy, and data governance standards

4.5 Design Output

Design outputs include:

- Integration Architecture Blueprint (showing NR–ReX system connectivity)
- Data Migration Mapping Document
- Standard Operating Procedures (SOPs) for unified scheduling and fleet management
- User and Technical Manuals
- Testing Reports and UAT Sign-offs

All design outputs will undergo version control, review, and approval before release to implementation.

4.6 Inspection and Review

Inspection and review processes ensure deliverables are complete and meet standards.

Inspection Activities:

- Internal peer review of all design and test documents.
- Technical Lead verification of integration design.
- UAT review with depot teams and Change Manager.
- Quality audit by the Project Manager before approval.

Defect Management:
All defects found during reviews will be logged in Jira and tracked until they are resolved.

4.7 Approval and Acceptance

All major outputs require two-level approval:

- Project Manager Approval: Confirms technical completion and documentation quality.
- Business Sponsor Acceptance: Confirms alignment with business objectives and operational readiness.

4.8 Authorisation and Distribution

After approval, the Project Manager authorizes the official release of deliverables.

Final versions are stored in the SharePoint Output Library with read-only access.
Distribution List:

- Project Team
- Steering Committee
- NR IT Department
- ReX Depot Managers (for operational documents)

Each released document is tagged with a version number, author, and approval date.

4.9 Updating and Changing

All outputs are subject to controlled updates through the formal Change Control Process

Update Process:

1. Change request submitted by team or stakeholder.
2. Impact analysis conducted by Project Manager.
3. Approval from Steering Committee.
4. Updated version released with new revision number and recorded in Output Register.

Updates may occur after pilot implementation or post-rollout to reflect lessons learned and new regulatory requirements.

5 Test Plan

The NR – ReX Network Integration Project consists in the superimposing of scheduling systems, fleet tracking devices, and data platform operation. The testing will verify that each system, process, and data flow, are working properly, efficiently and are secure, before the country-wide deployment. The plan outlines the testing methodology, procedures, roles and documentation necessary to ensure that all test execution delivers a product that meets functionality, performance and business expectations.

5.1 Introduction

The testing strategy follows a structured Waterfall approach, where each phase must pass verification before moving to the next.

The following levels of testing will be applied to all deliverables:

- Unit Testing – verify individual components (scripts, modules, interfaces).
- System and Integration Testing – ensure combined modules work together as one system.
- User Acceptance Testing (UAT) – confirm the system meets business needs and user expectations.

All tests will identify:

- the version of the component under test,
- the test specification version used, and
- the responsible tester and date.

Testing will be conducted using documentation prepared in the Project Plan (Stage 2) and approved by the Project Manager and Technical Lead.

The testing environment will mirror the real deployment setup to ensure accurate and reliable results.

5.2 Unit Testing

Each integration module, migration script, and dashboard component will undergo unit testing before being combined into the main system.

Unit testing verifies that each function performs as designed and handles data correctly.
Responsibilities

- Conducted by the Technical Lead and Integration Engineers.
- Reviewed and logged by the Project Manager for traceability.

Key Tests

- Data migration scripts correctly transform and load data.
- API connections between NR and ReX respond without errors.
- User-interface components (tracking dashboards) render correctly.
- Logging, error-handling, and exception routines function as intended.

All results will be recorded in the Testing Register stored in the SharePoint repository, clearly stating:

- Module Name,
- Version Number,
- Test Case ID,
- Expected Outcome, and
- Actual Outcome.

5.3 System and Integration Testing

5.3.1 Approach

Focus Areas

- Data flow validation between NR and ReX systems.
- System security testing – confirming that user roles and access levels are functioning.
- End-to-end transaction testing – validating route creation, vehicle updates, and reporting.
- Failover and recovery tests to ensure system stability.

Tools Used

- Jira for defect tracking.
- SQL test scripts for data validation.
- Microsoft Teams for coordination of test runs.

5.3.2 Review

After testing is done, results are reviewed by the Technical Lead, Business Analyst, and Project Manager.

A System Integration Testing Summary Report will document:

- Passed / Failed cases,
- Outstanding issues,

- Mitigation actions, and
- Sign-off recommendation for UAT entry.

5.4 Acceptance Testing

5.4.1 Approach

Objectives

- Validate that all functions align with approved business requirements.
- Ensure system performance, usability, and reliability are acceptable.
- Confirm operational readiness before the full rollout.

Activities

1. Use of real operational data to create UAT scenarios.
2. Conduct testing at limited depots
3. Record feedback and identify any defects/improvement reported.
4. Ensure that all necessary functions are checked that they all perform when needed.
5. Ensure that sign-off from the Business Sponsor is received after finishing.

5.4.2 Review

All acceptance-testing results will be documented in the UAT Report, which will include:

- Test scenario summaries,
- Pass/fail status,
- User feedback, and
- Corrective actions taken.

5.5 Certification of Test Results

After successful completion of all testing stages, the Project Manager will prepare a Testing Completion Certificate, summarising:

- Testing phases completed,
- Evidence of compliance with design specifications,
- UAT sign-offs by NR and ReX representatives, and
- Final acceptance by the Business Sponsor.

Each certificate will be signed by:

- Project Manager (Kavish Gunasekara) – certifying completion;
- Business Sponsor (Michael) – confirming business acceptance; and
- Executive Sponsor (Lecturer) – confirming academic and governance approval.

All documentation, including test cases, logs, and acceptance forms, will be archived in the Project Records Management System for future reference and audit purposes.

6 Implementation and Delivery Plan

This section describes how the NR–ReX Network Integration Project will be implemented, transferred, and handed over to NR and ReX management.

The implementation phase is where the Project Team delivers the final integrated systems to the Business Owners and they can experience all operational benefits of integration such as improved coordination, efficiency, and real-time visibility across the network.

6.1 Implementation

Implementation is the most important stage in the project lifecycle, where the developed and tested product is used for live use and the responsibility shifts from the Project Team to the Business Owners.

This phase ensures that all operational, technical, and organizational aspects are ready to be used, adopted and used successfully.

Implementation Objectives

- Deliver and activate the integrated NR–ReX scheduling and tracking systems.
- Ensure business continuity and minimal disruption during rollout.
- Transfer ownership of project deliverables to the Business Owners.
- Train and support operational staff during the transition period.
- Establish clear monitoring and maintenance procedures post-handover.

Implementation Responsibilities

Role	Key Responsibilities
Project Manager (Kavish Gunasekara)	Oversees all deployment activities, coordinates between technical and operational teams, and ensures quality assurance compliance.
Business Analyst (Sarik)	Supports end-user documentation, verifies that business processes align with the new system, and monitors adoption rates.
Technical Lead (Darren)	Leads technical deployment, ensures stable system installation, and manages data migration verification.
Change Manager (NR Staff)	Coordinates communication with staff, manages change readiness, and organises post-rollout training.
Business Sponsor (NR COO)	Provides strategic oversight, approves the go-live transition, and confirms operational readiness.

Step-by-Step Implementation Approach

1. Pre-Implementation Preparation
    - Confirm completion and approval of all testing (UAT sign-off).

    - Finalise training sessions for depot and scheduling staff.
    - Back up all legacy system data.
    - Prepare the implementation environment (hardware, servers, access control).

2. Pilot Site Deployment
    - Begin deployment at selected depots.
    - Monitor performance and address early issues.
    - Gather user feedback and fine-tune configurations.

3. Full National Rollout
    - Sequentially deploy systems to remaining NR and ReX depots.
    - Provide on-site technical and training support.
    - Maintain daily progress and issue logs.

4. Handover and Post-Implementation Support
    - Transfer all system ownership to NR operations.
    - Document and confirm acceptance of deliverables.
    - Provide 3 months of post-rollout support, addressing any technical or user issues.
    - Conduct a post-implementation review to evaluate project success and lessons learned.

Resources Required

- Human Resources: Technical Lead, Integration Engineers, Trainers, and Change Manager.

- Infrastructure: backup systems, Secure servers, network stability and VPN access.

- Support Tools: Jira (issue tracking), SharePoint (documentation), Microsoft Teams (coordination)

- Training Facilities: Digital platforms and Workshops for user engagement.

Outcome Realization and Transition

Following the rollout the business Owners of the NR and ReX management businesses will have full control and operation.

The emphasis will move from the ability to provide to actually achieving benefits such as increased efficiency, improved customer satisfaction and lowered logistics costs, and these benefits will be measured using performance metrics.

A formal Outcome Realisation Plan will later evaluate:

- Achievement of project goals and KPIs.
- Measured improvements in delivery times and cost savings.
- Continued staff competency and adoption rates.

7 Project Evaluation Review(s)

The Project Evaluation Reviews are structured assessments conducted throughout and after the project lifecycle to measure performance, confirm success criteria, and identify improvement opportunities.

Timing of Reviews

Review Stage	Timing	Purpose
Phase Review	At the end of each major milestone (M1–M7)	To verify that objectives, deliverables, and schedules are on track before proceeding to the next stage.
Post-Implementation Review (PIR)	One month after full rollout (Month 20)	To evaluate project outcomes, financial performance, and system functionality in live operations.
Lessons Learned Review	Within two months after project closure	To capture feedback, document best practices, and identify improvement areas for future projects.

7.1 Scope of Reviews

Multiple dimensions were assessed in the evaluation of the projects. A technical review was performed to validate that the specifications, performance, security and best practice for systems, technologies and integrations were used in the solution. The process review addressed the effectiveness of project management, risk management techniques and communication throughout the implementation. Measures of success included quantity of savings or efficiency gains, and level of staff uptake, among others, but made a judgment/comparison based on those factors versus what the original goal of cost savings or efficiency gain was. The budget report was a financial review of all actual costs that were incurred compared to what was budgeted in efforts to access return on investment and budget control. Finally, the stakeholder review gathered feedback from both National Railway (NR) and ReX teams, including managers and users, to assess satisfaction and identify any remaining issues.

7.2 Review Participants

Role	Responsibility
Project Sponsor (Lecturer)	Approves final evaluation reports and ensures alignment with project objectives.
Business Owner (NR COO)	Reviews operational performance and validates business value realization.
Project Manager (Kavish Gunasekara)	Coordinates review sessions, compiles performance data, and leads the lessons-learned workshop.

Business Analyst (Sarik)	Collects operational data, user feedback, and compares performance metrics to baseline.
Technical Lead (Darren)	Provides technical documentation, testing results, and post-rollout system analysis.
Change Manager (NR Staff)	Gathers staff and user feedback, adoption rates, and training impact reports.

7.3 Post-Implementation Review (PIR) Process

The PIR review will come into effect after the project has concluded and has output given over and is able to be assessed. The review will evaluate if the project achieved its performance targets and desired outcomes. It considers the comparison of planned vs. actual project results, data accuracy, financial performance metrics and system uptime alongside other forms of information. The results of the review will be put into a PIR report, ready to be presented to the Steering committee and Business Sponsor.

7.4 Lessons Learned and Continuous Improvement

The topics covered will include:

- Effectiveness of project communication and collaboration tools.
- Performance of risk and change management strategies.
- Resource utilization and budget accuracy.
- Technical challenges and their resolutions.
- Recommendations for similar integration projects in the future.

All lessons will be compiled into the Project Evaluation Report, stored in the organization's knowledge base for reference in future projects.

7.5 Review Reporting and Follow-Up Actions

The each review will result in an official report, including:

- Summary of findings
- Recommendations for improvement
- Assigned responsibilities for follow-up actions
- Target deadlines for implementation

The reports will be distributed to:

- The Steering Committee
- The Business Sponsor
- The Project Team
- Relevant NR operational departments

7.6 Post-Project Evaluation Ownership

They will also monitor key performance indicators such as cost savings, uptime and

efficiency. Quarterly operational reports will also be created during the first-year post-implementation to track progress. Feedback from system users and depot personnel will also be utilized to improve future updates and reviewed from time to time and processes.

8 Project Plan

The project plan explains the scheduled baseline of the project, interdependent relationships, and milestones. All activities will be clearly defined, tracked for the 20-month duration of the project and resourced. It is used as a reference to monitor performance, change control and report progress.

The Project Plan is linked with the Project Charter which was in stage 1 and Project Plan Document which was in stage 2 to show updates made during execution planning.

This section explains the framework of the plan and how it is kept up-to-date to help in delivering project outcomes.

8.1 Overall Project Plan

Purpose
The Project Plan provides a single, approved schedule against which progress, risks, and performance are measured. It identifies the sequencing of work, resource utilisation, and interdependencies between tasks to achieve the project's objectives within scope, time, and budget.

Key Components

The Project Plan includes:

- Work Breakdown Structure (WBS) – Divides the project into seven key phases (Initiation to Closure) with assigned tasks and responsibilities.
- Timeline and Milestones – A 20-month roadmap highlighting key checkpoints (M1–M7) for approval and progress tracking.
- Resource Allocation – Identifies team roles, time commitments, and tool usage (Microsoft Project, Jira, SharePoint).
- Dependencies and Deliverables – Defines task relationships and completion criteria for each output.
- Performance Baseline – Provides the foundation for variance analysis in cost, schedule, and risk monitoring.

Plan Maintenance

- The Project Manager (Kavish Gunasekara) is responsible for maintaining the day-to-day working copy of the Project Plan in Microsoft Project and SharePoint.

- The Steering Committee will review changes such as alterations to timelines, dependencies or resources in the Change Control Register, at milestone meetings.

- The baseline plan in Appendix A will only be revised following formal approval from the Business Sponsor and Executive Sponsor to maintain document integrity.

Monitoring and Reporting

Project progress will be reviewed through:

- Weekly meetings for tracking tasks.

- Monthly regular reports involving finalizing milestone progress, risks, and financial updates.

- Milestone reviews (M1–M7) requiring formal sponsor sign-off before advancing to the next phase.

- Performance metrics evaluated against the criteria defined in Stage 2 (Section 8 – Performance and Metrics).

Tool Utilization
To ensure real-time project visibility and consistency:

- Microsoft Project – used for scheduling, dependencies, and resource assignments.

- SharePoint / OneDrive – document management, version control, and collaboration.

- Jira / Asana – issue tracking and sprint management (for integration and testing tasks).

- MS Teams – meetings, communication, and coordination between NR and ReX teams.

Change Control and Versioning

- Minor updates (e.g., task progress) are documented/recorded in regular status reports.

- Major revisions (e.g., re-phasing of activities or resource reallocations) require formal approval and a new baseline version.

- Version control will be tracked using standard document numbering (e.g., V1.0 – Initial Baseline, V1.1 – Post-Design Update, V2.0 – Post-Rollout).

9 Appendices

9.1 Appendix A – Project Plan

Milestone	Description	Target Month
M1	Project Initiation Approved	Month 1
M2	Requirements and Analysis Complete	Month 4
M3	Design Approved	Month 6
M4	Build and Data Migration Completed	Month 12
M5	Pilot Evaluation Complete	Month 15
M6	Full Integration Complete	Month 19
M7	Project Closure Approved	Month 20

9.2 Appendix B – Risk Analysis

Risk	Impact	Likelihood	Mitigation Strategy
System integration delays	High	Medium	Conduct early testing and vendor checkins
Data quality issues	High	High	Data cleansing and validation before migration
Staff change resistance	Medium	High	Provide training and clear communication
Vendor API delays	Medium	Medium	Escalate through Service Level Agreements
Budget overrun	High	Medium	Implement cost tracking and contingency fund

9.3 Appendix C – Budget and Expenditure Summary

Category	Allocated Budget (AUD)	% of Total
Personnel & Labour	515,000	60.6 %
Software & Licensing	70,000	8.2 %
Hardware & Infrastructure	65,000	7.6 %
Training & Change Management	55,000	6.5 %
Testing & Quality Assurance	35,000	4.1 %
Operations & Travel	25,000	2.9 %
Contingency Reserve	85,000	10 %
Total	850,000	100 %

9.4 Appendix D – Environmental Baseline

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Environment Component	Description
Software	Microsoft Project, SharePoint, MS Teams, Jira
Hardware	Secure servers, VPN access, backup systems
Development Environment	Windows Server environment with secure cloud integration
Testing Environment	Dedicated sandbox for UAT and integration testing
Communication Infrastructure	Microsoft Teams and email for collaboration

9.5 Appendix E – Stakeholder Agreements

Stakeholder	Role	Commitment / Responsibility
Executive Sponsor (Ochini)	Project Oversight	Approves major milestones and final deliverables
Business Sponsor (NR COO)	Funding & Strategic Alignment	Ensures alignment with NR's objectives
Project Manager (Kavish Gunasekara)	Project Delivery	Leads planning, execution, and monitoring
Business Analyst (Sarik)	Requirements & Testing	Captures business needs and validates system functionality
Technical Lead (Darren)	System Integration	Designs architecture and oversees technical testing
Change Manager (NR Staff)	Training & Adoption	Supports user training and change management

9.6 Appendix F – Responsibility Matrix (RACI)

Task / Phase	PM	BA	Tech Lead	Change Mgr	Sponsor
Project Initiation	A	R	C	I	A
Requirements & Analysis	A	R	C	I	A
System Design	A	C	R	I	A
Build & Integration	A	C	R	I	A
Pilot Implementation	A	C	R	R	A
Full Rollout	A	C	R	R	A
Closure & Handover	A	C	C	I	A

Legend:

- R – Responsible (performs task)
- A – Accountable (final decision authority)
- C – Consulted (provides input)
- I – Informed (kept updated on progress)

9.7 Appendix G – Definitions

Term	Definition
PEP	Project Execution Plan – operational document for project delivery
WBS	Work Breakdown Structure – hierarchical decomposition of tasks
UAT	User Acceptance Testing – testing conducted to validate requirements
SLA	Service Level Agreement – commitment between vendor and client
Baseline	Approved version of project plan used for tracking and control

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